

CMPE2004 Advanced Programming

Term Project Assignment

Android Studio Mobile Application Project for Disadvantaged Individuals

Course: CMPE2004 Advanced Programming

Semester: 2025–2026 Spring

Instructor: Dr. Günet Eroğlu

Project Theme

For the term project, students will develop an **Android mobile application in Java using Android Studio for disadvantaged individuals**. The aim is to design a useful, accessible, and socially beneficial mobile app that helps users in daily life.

The project must reflect the programming and Android development topics covered in the course, including Java syntax, object-oriented programming, user interface design, activities, intents, navigation, internet usage, notifications, styles, themes, and Material Design.

Project Format

- The project will be developed in **Android Studio**
- The programming language will be **Java**
- The project can be completed **individually or in groups**
- Students must choose **one** of the following **three project options**

Project Options

Option 1: Accessible Daily Reminder App

Design an Android application for individuals who need support in remembering daily tasks, such as medication times, appointments, drinking water, or basic routines.

Example features:

- Add reminder titles and times
- Display daily reminder list using simple layouts
- Notification alerts
- Easy navigation between screens
- User-friendly interface with large buttons and readable text

Option 2: Emergency Help and Quick Contact App

Design an Android application for disadvantaged users who may need quick communication with caregivers, relatives, or emergency contacts.

Example features:

- Emergency contact list

- One-tap emergency call buttons
- Screen for important personal information
- Intents for call or message actions
- Notification or alert support
- Simple and accessible design

Option 3: Supportive Information App

Design an Android app that provides helpful information for disadvantaged individuals, such as health tips, legal rights, social support services, transportation help, or education support.

Example features:

- Category-based information screens
- Internet connection to retrieve simple online information
- Menus for navigation
- Accessible design with clear text and icons
- Multiple screens and structured content

Minimum Requirements

Each project must include the following:

1. User Interface Components

The app must use basic mobile UI components such as:

- TextView
- EditText
- Button
- ImageView
- ScrollView
- Menus

2. Multiple Screens

The application must include **at least 3 different activities/screens**.

3. Activity Lifecycle

Students should demonstrate awareness of the Android activity lifecycle in their app design.

4. Intents and Navigation

The app must use:

- Explicit intents between activities

- Menus or buttons for navigation
- At least one practical Android intent where appropriate

5. Object-Oriented Programming

Students must demonstrate:

- Class and object usage
- Encapsulation
- Inheritance or interfaces where appropriate
- Exception handling
- Clean code structure

6. Internet / HTTP Feature

The app must include at least one internet-related function, such as:

- Retrieving data from a simple web source or API
- Displaying online information
- Sending a basic HTTP request

7. Notification / Broadcast / Service

The app must include at least one of the following:

- Notification
- Broadcast receiver
- Service

8. Design and Visual Quality

The app should include:

- Styles
- Themes
- Material Design principles
- Simple animation or visual improvement

9. Data Rule

From internet, user server apps.

Deliverables

Students must submit:

1. Source Code

A complete Android Studio project.

2. Report

A short report in English including:

- Project title
- Student name(s) and ID(s)
- Selected project option
- Purpose of the app
- Main features
- Technical structure
- Challenges and solutions
- Conclusion

3. Presentation

Students will present their project during **Week 14**. The presentation should include:

- Problem definition
- Why the selected user group is important
- Main app features
- Screens and demo
- Technical explanation

Evaluation Criteria

The project will be graded as part of the course project component.

Grading Breakdown

- Functionality and completeness – 25 points
- Use of Java and OOP – 15 points
- User interface and usability – 15 points
- Activities, intents, and navigation – 15 points
- Internet / notification / service features – 15 points
- Code quality and organization – 10 points
- Report and presentation – 5 points

Academic Integrity

All work must be original. Copying code, plagiarism, or submitting another person's work is strictly prohibited and will be evaluated according to university academic integrity rules.

Final Note

Students are expected to create an app that is not only technically correct but also socially meaningful, accessible, and helpful for disadvantaged individuals.